① Bitwise AND(&)

The bitwise AND operator compares each bit of two numbers. If both bits are 1, the corresponding result bit is 1; otherwise, it is 0.

* Binary of 4:0100
* Binary of 2:0010
* Result of 4&2:0000
* Final answer:0

Bitwise OR(|)

The bitwise OR operator compares each bit of two numbers. If either bit is 1, the corresponding result bit is 1; otherwise, it is 0.

* Binary of 4:0100
* Binary of 2:0010
* Result of 4|2:0110
* Final answer:6

* Bitwise XOR(^)

The relational operators compare two values and return a boolean result bit is 1; otherwise, it is 0

* Binary of 4:0100
* Binary of 2:0010
* Result of 4^2:0110
* Final answer: 6

② Relational operations

The relational operators compare two values and ~~te~~ return a boolean result (true or false). Assuming 1 represents true and 0 represents false

* 4<2: False(0)
* 4>2: True(1)
* 4==2: False(0)
* 4≥2: True(1)
* 4≤2: False(0)
* 4!=2: True(1)

③
i) Here are some multiple-choice questions (MCQs) about increment and decrement operators in programming. The key concepts involve understanding the difference between prefix (++a) and postfix (a++) increment/decrement, and how they behave in expressions.

ii) The term "right and left operations" must commonly refers to bitwise shift operations (<<, >>, >>>) in programming or array/string rotation/shift operations in data structures and algorithms. the following MCQs cover bitwise operations, which are a highly probable subject for this query.

Questions:-

① Increment and decrements:-

1) what will be the output of the following c program?

```c
#include <stdio.h>
int main() {
int x = 5;
int y = x++ / 2;
printf("%d", y);
return 0;
}
```

a) 3

b) compile - time error

c) None of these

d) 2

Answer: d) 2.

29Q what will be the output of the following c program

```c
#include <stdio.h>
int main() {
int a=4, b, c;
b=--a;
c=a--;
printf("%d %d", a, b, c);
return 0;
}
```

a) 332

b) 232

c) 322

d) 233

Answer: d) 233

30Q what will be the output of the following c code?

```c
#include <stdio.h>
int main() {
int i=0;
int x= i++, y= ++i;
printf("%d %d\n", x, y);
return 0;
}
```

a) 0, 2

b) 0, 1

c) 1, 2

d) undefined

Answer: a) 0, 2

② MCQ's of right & left operations

1. what is the result of the expression 5<<2?

a) 10

b) 20

c) 1

d) 2.5

* Answer b) 20

2. The right shift operator >> is equivalent to which arithmetic operation for positive integers?

a) Multiplication by 2^n

b) Division by 2^n

c) Addition of 2^n

d) subtraction of 2^n

* Answer b) Division by 2^n

3. when a signed integer with a negative value is right-shifted (>>), how are the vacant leftmost bit positions generally filled in many programming languages like c/c++ or java?

a) with 0s (zero-filled)

b) with 1s (sign bit extension)

c) with the result of a modulo operation

d) the behavior is undefined / implementation-defined.

* Answer:- b) with 1s (sign bit extension) OR d) the behavior is undefined / implementatio -defined.//